Topology	Node degree	Diameter	Bisection width	Arc connectivity	Cost	Suitability for EV Charging WSN	Reasoning
Linear Array	1 or 2	N − 1	1	1	N − 1	No	Simple and inexpensive, but represents a 1D structure. Its O(N) diameter and low fault tolerance make it unsuitable for a geographically distributed 2D charging network
Ring	2	N/2	2	2	N	No	Provides an alternative path compared with a linear array, but each node only has two neighbours and the topology remains fundamentally 1D. It does not accurately represent the 2D Cartesian grid.
Star	1 or N − 1	2	1	1	N − 1	No	Offers short paths, but all communication depends on a central node rather than with its neighbouring charging nodes. This creates a bottleneck and single point of failure and does

							not naturally support neighbour-to-neighbour communication.
Binary Tree	1, 2 or 3	2log((N+1)/2)	1	1	N − 1	No	Efficient hierarchical structure with a relatively small diameter, but the parent-child hierarchy does not represent the geographical neighbour relationships between charging stations, and most importantly, its arc connectivity is only 1.
2D Mesh	2, 3 or 4	$2(N^{1/2} - 1)$	$N^{1/2}$	4	$2(N - N^{1/2})$	Yes — Selected	Directly represents the 2D Cartesian/grid arrangement in the specification. Each internal node can communicate with up to four immediately adjacent neighbours, matching the WSN's neighbour-query requirement. It provides multiple paths and scales naturally as the network expands.
2D Wrap-Around	4	$N^{1/2}$	$2N^{1/2}$	4	2N	Partially	Provides shorter paths and uniform connectivity, but wrap-around links create artificial connections between

							geographically distant boundary nodes. This reduces physical realism compared with a standard 2D mesh.
3D Cube	3, 4, 5 or 6	$3(N^{1/3} - 1)$	$N^{2/3}$	3	$2(N - N^{2/3})$	No	Provides good connectivity and fault tolerance, but the additional spatial dimension is unnecessary because the charging nodes are modelled on a 2D Cartesian plane.
Hypercube	log N	log N	N/2	log N	(N log N)/2	No	Has excellent parallel-computing properties, including O(log N) diameter, but does not naturally represent geographical proximity and is less suitable for an arbitrary charging-station layout.
Completely connected	n-1	1	$N^2/4$	N-1	$N(N - 1)/2$	No	Every node can communicate directly with every other node, but the $O(N^2)$ connection cost becomes excessive as the WSN grows. It also does not reflect the required immediate-neighbour

							communication pattern.